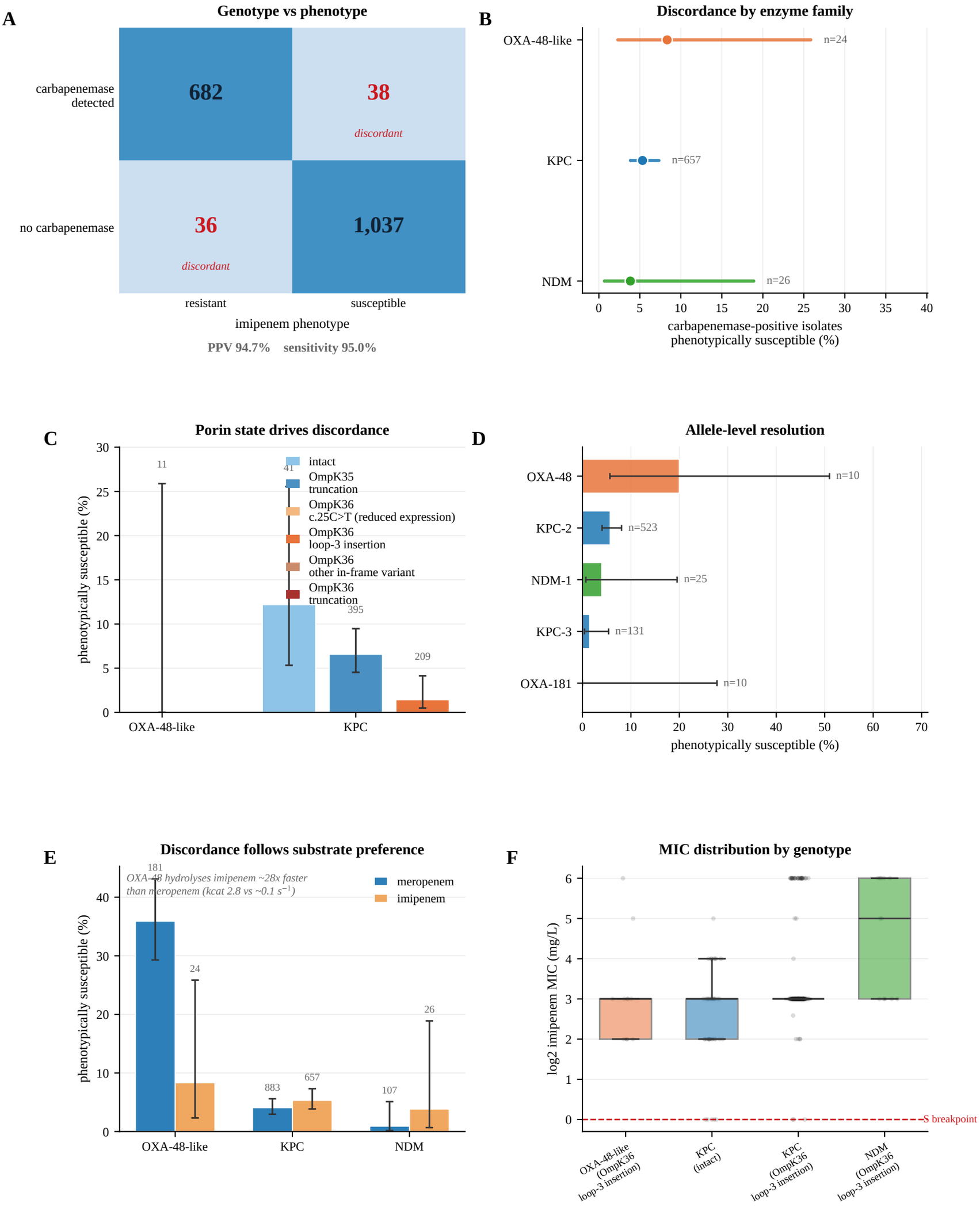
A
Genotype vs phenotype
carbapenemase detected
no carbapenemase
682
38
discordant
36
discordant
1,037
resistant
susceptible
imipenem phenotype
PPV 94.7%
sensitivity 95.0%
B
Discordance by enzyme family
OXA-48-like
KPC
NDM
n=24
n=657
n=26
0
5
10
15
20
25
30
35
40
carbapenemase-positive isolates phenotypically susceptible (%)
C
Porin state drives discordance
phenotypically susceptible (%)
intact
OmpK35 truncation
OmpK36 c.25C>T (reduced expression)
OmpK36 loop-3 insertion
OmpK36 other in-frame variant
OmpK36 truncation
11
41
395
209
OXA-48-like
KPC
D
Allele-level resolution
OXA-48
KPC-2
NDM-1
KPC-3
OXA-181
n=10
n=523
n=25
n=131
n=10
phenotypically susceptible (%)
E
Discordance follows substrate preference
OXA-48 hydrolyses imipenem ~28x faster than meropenem (kcat 2.8 vs ~0.1 s⁻¹)
meropenem
imipenem
phenotypically susceptible (%)
181
24
883
657
107
26
OXA-48-like
KPC
NDM
F
MIC distribution by genotype
log2 imipenem MIC (mg/L)
S breakpoint
OXA-48-like (OmpK36 loop-3 insertion)
KPC (intact)
KPC (OmpK36 loop-3 insertion)
NDM (OmpK36 loop-3 insertion)